# Drift to Default: How Autoregressive Video Rollouts Forget Their Opening

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**Abstract**

Autoregressive video diffusion models generate long videos chunk by chunk and drift over tens of seconds. We measure drift by intervention: we shift the brightness, colour or scene class of the same opening frame and measure how much of the shift survives. In 23 preregistered studies on nine open models (1.3B to 14B parameters, four architectures), rollouts *relax*: global tone and scene content move from the opening toward a default set by the conditioning, keeping a share $\alpha$ that depends on the model, the scene and the horizon. Under a generic city prompt, openings of six scene classes lose 39–93% of their class evidence within 30 s. What decides the long-run share is how the opening is held. An opening seen only through a sliding key–value window is forgotten, at 1.3B and at 14B parameters. Re-denoising overlapping frames delays forgetting by about 20 s but does not stop it. An opening kept next to the window at the positions the model was trained on holds a steady share, about 0.2 for an attention sink of any depth and half for a clean copy in every section. A sink left at its absolute rotary positions keeps 0.6, mostly through periodic bursts in which the opening, including its scene content, returns. [study 23: whether $\alpha$ is independent of the size of the shift.] A linear chunk recursion predicted the prompt's per-chunk pull (0.08); bit-exact cache replay measured 0.088 $[0.045, 0.149]$, but the recursion holds only for about 20 s. In an exploratory analysis, drift separates into forgetting the opening and an offset of the default from where the same prompt starts, near zero only in models trained on their own long rollouts. A content-aware correction and a deeper sink both reduce drift without training. We report all 78 preregistered primary tests, N of which failed.

## 1 Introduction

Autoregressive video diffusion models generate a video chunk by chunk, each chunk conditioned on the model's own earlier output [3, 1, 2]. This makes streaming and open-ended generation possible and is the basis of interactive world models [38, 39, 40]. It also makes them drift: over tens of seconds the picture darkens or saturates, loses texture, and its content wanders. A growing family of methods targets this failure with rolling windows, attention sinks, re-aligned positions, longer self-rollouts and test-time correction [7, 8, 9, 22, 12, 13, 21]. Most report a few samples per prompt and an aggregate quality score. What drift *is*, in quantitative terms, is rarely measured.

We measure it by intervention. We hold an opening frame, shift its brightness, colour or scene content *within* a scene, let the model continue, and ask how much of the shift survives. The answer has the same form in every model we tested: the rollout *relaxes*. It moves from its opening toward a default state set by the conditioning, and keeps a model-specific share $\alpha$ of where it started,

$$x_\infty \approx \alpha\, x_{\text{open}} + (1-\alpha)\,\mu. \tag{1}$$

Written this way the law is just a linear response, and much of the paper is about which parts of it carry content. Three findings do.

**What survives in the long run depends on how the opening is held, not on autoregression or scale** (Fig. 4). An opening seen only through a sliding key–value window is forgotten, in 1.3B models and in a 14B model alike. Re-denoising overlapping frames passes the opening forward and delays forgetting by about 20 s. An opening kept next to the window, at positions the model was trained on, holds a steady share: about 0.2 for an attention sink whatever its depth, half for a clean copy in every section. A sink left at its absolute rotary positions keeps more, but mostly through periodic bursts in which the opening returns.

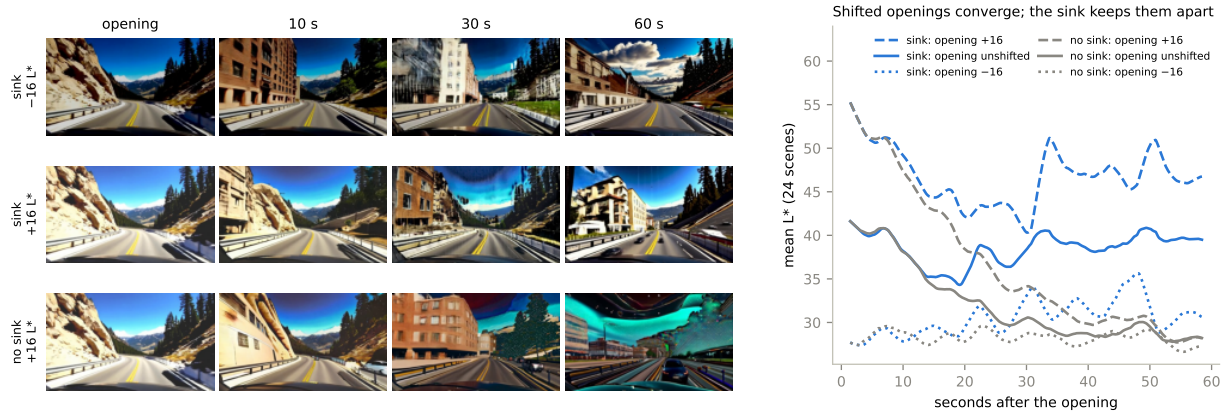



**Figure 1:** Relaxation. Left: a generated mountain-road opening, darkened or brightened by 16 L* before encoding, continued for 60 s under a generic city prompt by Self Forcing DMD with and without an attention sink. The scene turns into a city street in all three runs; without a sink the colours also blow out. Right: mean brightness over the 24 scenes of study 7 for the three versions of each opening (3-s moving average). Without a sink the versions converge to one default; with a sink they stay apart by a share that fluctuates around half.

**The default is set by the conditioning.** Lighting words move the default brightness by up to 27 L*, and scene words decide whether a snowy, desert or forest opening stays or becomes a city street within 30 s. Which parts of the conditioning carry tone and which carry the scene differs between models.

**Drift has two separable parts** (exploratory): forgetting the opening, $(1-\alpha)(x_{\text{open}}-\mu)$, and an offset of the default itself from where the same prompt starts, which is small in the two models trained on their own long rollouts and 4–20 L* in the others.

A linear chunk recursion, $\alpha_\infty = w_s/(w_s+w_p)$ with per-chunk pulls of the sink and the prompt, predicted $w_p \approx 0.08$; bit-exact replay of the key–value cache measured 0.088. It describes the first 20 s, not the positional effects that follow. Training-free levers act on either term: an early-step tone correction, a deeper sink, and lighting words calibrated to a model's measured defaults, each with its own cost.

**Contributions.**

1. A measurement protocol for long-horizon drift: held and perturbed openings, persistence $\alpha$ and default $\mu$ for tone and scene content, paired seeds pinned to one GPU architecture, and preregistered analyses.
2. An empirical account of relaxation across nine open models (six of the Wan2.1-1.3B architecture, the Wan2.1-14B student Krea Realtime, the HunyuanVideo-based FramePack and MAGI-1), for tone and scene content, at horizons up to 120 s: what sets $\mu$, what sets the long-run share (how, and at which position, the opening is held), and an exploratory decomposition of drift into forgetting and attractor offset.
3. A phenomenological recursion whose predicted prompt pull we preregistered and then measured by bit-exact replay, with its limits.
4. Training-free levers matched to the two terms, compared on the same prompts and seeds, with their costs and failures.

We preregistered 23 studies with hashed analysis code and report all 78 primary tests, including the N that failed (Table 3). No model is trained; all code and per-video statistics are public.

## 2 Related work

**Autoregressive and long video diffusion.** Diffusion Forcing [3] and History-Guided Video Diffusion [33] enable causal rollout with per-frame noise and guidance on the history. CausVid [1] distils a bidirectional teacher [4] into a few-step causal student with DMD [5]; Self Forcing [2] and Self-Forcing++ [22] train on the

student's own (minute-long) rollouts. LongLive [7], Rolling Forcing [8] and Deep Forcing [9] add attention sinks, rolling joint denoising and deeper, re-aligned sinks; LoL [21] traces "sink collapse" to positional periodicity. Segment-wise models condition on overlapping frames [10, 26, 31], a temporal pyramid [30] or packed history [23]; training-free schemes lengthen bidirectional models [25, 27, 28, 29]. These works propose mechanisms against drift; we measure what drift does and which mechanism changes which part of it.

**Error accumulation, sinks and correction.** Sampling from one's own output causes exposure bias in sequence models [34, 35] and diffusion models [36, 37]; error analyses of autoregressive video diffusion [14] bound how errors compound, and world models [38, 39, 40, 50] add noise to context frames for robustness. For global statistics our results suggest a complementary view: the rollout forgets its context and relaxes to the conditioning's default. Global brightness biases of diffusion models arise at the earliest denoising steps [41, 42]. Streaming language models keep the first tokens as an attention sink [11, 43], related to registers in vision transformers [44]; the closest analogue to our finding is instruction drift in dialogs [45], where attention to the system prompt decays and behaviour reverts to a default. Pathwise test-time correction [12] and FreqForcing [13] anchor rollouts to early frames or their low frequencies; our correction is a minimal member of this family. VBench [15] scores imaging quality with MUSIQ [16]; we measure drift against each prompt's fresh-start distribution [17] and by intervention.

# 3 Measuring drift by intervention

**Models.** We study nine open models (Appendix Table 2). Self Forcing (five checkpoints), CausVid, LongLive and Rolling Forcing are causal few-step students of Wan2.1-T2V-1.3B [4] that read their past through a key–value cache of 21 latent frames (LongLive: 12), optionally with an attention sink (the first chunk kept in the cache); Krea Realtime 14B [32] is a student of the 14B model. SkyReels-V2-DF [10] (same architecture as the 1.3B model) extends videos in overlapping segments, Matrix-Game-2.0 [50] is a Wan-architecture world model driven by actions, FramePack [23] (13B, from HunyuanVideo [24]) conditions every section on the opening, and MAGI-1 [31] (4.5B) has its own VAE and transformer; Section 5 describes how each holds the opening. All Wan-1.3B cache configurations run in one harness that reproduces the official Self Forcing loop bit for bit when no intervention is applied.

**Held and perturbed openings.** An opening image is held still for the first chunk (9 frames, 0.56 s at 16 fps; SkyReels: 17 frames as a prefix video; FramePack and Krea: the input image; MAGI-1: a clean image prefix) and continued under a prompt. Openings are NAVSIM [19] front-camera frames or frames generated by the models themselves. To separate what the model keeps from what the scene is, we shift the *same* opening in CIELAB before encoding: $L^*$ by $\pm 8$ and $\pm 16$, $b^*$ by $\pm 12$. For scene content, we instead generate openings of six scene classes and vary the prompt (Section 4.1).

**Persistence and default.** For a statistic $x$ (a per-frame CIELAB mean), the persistence at time $T$ is the within-scene least-squares slope of the end value $\bar{x}(T)$, averaged over the window $[T-w, T)$ after the opening, on the opening value $x_{\text{open}}$ (mean of the held frames),

$$\alpha(T) = \frac{\partial\, \bar{x}(T)}{\partial\, x_{\text{open}}}\Big|_{\text{scene}}, \qquad \mu(T) = \frac{b_0}{1-\alpha(T)}, \tag{2}$$

where $b_0$ is the fitted intercept, so $\mu$ is the level at which the end equals the opening (the zero-drift level). The window is the last 2 s at 31 s (study 6), 5 s at 20–30 s (studies 8, 9, 11), 10 s at 60 s and at 30 s in studies 12 and 15, and 1 s for curves; Appendix Table 4 compares all models over the same window, 15–20 s. Averaging slopes over 24 scenes gives a scene-bootstrap interval. The fresh default is where prompt-only rollouts end.

**Content persistence.** For scene content we score each frame with CLIP ViT-L/14 [48] as the log-odds of the opening's class against "a city street", $z = s\,(\cos(f, t_{\text{class}}) - \cos(f, t_{\text{city}}))$, and define $\kappa(T) = (z(T) - \mu_z)/(z_{\text{open}} - \mu_z)$ with $\mu_z$ the class score of fresh rollouts of the generic prompt. $\kappa$ is the content analogue of $\alpha$: 1 if the class evidence is kept, 0 if the rollout looks like a fresh generic rollout.

**Drift metrics for prompt-only rollouts.** Where no opening is held, drift is measured against fresh 5-s generations of the same prompt, in colour statistics ($D$) and DINOv2 [17] features ($K$; Appendix B).

**Pairing and preregistration.** With bf16 kernels, the same seed and code give bit-identical videos within a GPU architecture but different videos across A40, A100/A800 and L40S/RTX 6000 Ada. Arms that are compared therefore run in one job, which deletes and regenerates its outputs if a preempted job resumes on another architecture. (In our first study this was not yet enforced; 62% of its nominal pairs crossed architectures, which widens but does not bias its intervals.) Each study's hypotheses, decision rules and analysis code were frozen by SHA-256 hash before its videos were generated (study 1: before analysis, after generation had started); studies 3–23 were also committed publicly first. Tests within a study are Holm-corrected. Table 3 lists every primary test.

# 4 Rollouts relax toward a conditioning-set default

**Tone.** Observationally, the 31-s change of every global statistic is anti-correlated with its opening value ($r$ from $-0.79$ to $-0.95$; Appendix Table 5), but openings differ in content too, so we intervened: we shifted the brightness and colour temperature of the same opening within each of 24 scenes (12 NAVSIM, 12 generated) and continued every version with the generic prompt (study 6, 31 s; study 10, 60 s). The versions converge (Fig. 1). At 60 s every cache configuration has relaxed (Fig. 2a): $\alpha$ ranges from $-0.01$ (LongLive without a sink) to 0.79 (Self Forcing GAN), ranked as at 31 s (Spearman $\rho = 0.78$, $p = 0.008$). Part of the loss is immediate: most configurations lose a fifth to a third of the shift within 4.5 s, and SiD loses almost all of its total there (0.66 of 0.65), which looks like the model rejecting an off-manifold shift rather than forgetting it. Colour temperature behaves like brightness. Persistence varies by checkpoint even with architecture, context and sampler fixed (Self Forcing: 0.35 to 0.79 at 60 s; two SiD checkpoints differ by 0.36), so it is a property of the checkpoint rather than of the training objective.

**The response is linear in the shift.** Equation 1 treats $\alpha$ as independent of how far the opening was shifted, but every $\alpha$ above comes from one pair of shifts ($\pm 16$ L$^*$). Study 23 shifted the same 24 openings by $-24$, $-8$, 0, $+8$ and $+24$ L$^*$ (Self Forcing, 60 s, two seeds, 720 videos) and tested equivalence within $\pm 0.15$. [results: large vs small shift with sink, re-rotated sink; darkening vs brightening; no sink forgets a large shift too.] In study 6, persistence from $\pm 8$ and from $\pm 16$ shifts agrees within 0.09 in 12 of 14 configurations at 31 s (exploratory; Appendix C).

## 4.1 Scene content

Global tone is the easiest statistic to shift and measure, but it is not what a user sees as "the scene". We therefore generated openings of six scene classes (snow, desert, forest, coast, mountain, farmland; four each, selected by CLIP before any continuation) and continued each under the generic city prompt and under its own class prompt, with eight configurations (study 9). Content relaxes like tone (Fig. 3). Under the city prompt, $\kappa$ at 30 s ranges from 0.07 (SiD) to 0.61 (Rolling Forcing), and the opening's class is CLIP's top label in only 34% of Self Forcing frames at 30 s. Under the opening's own class prompt the class persists ($\kappa$ 0.71–1.05; 9.3 [8.1, 10.5] more log-odds than under the city prompt, preregistered), as a fresh rollout of that prompt would also show it; the informative result is the loss under the city prompt. Content and tone persistence are correlated across configurations ($\rho = 0.67$, $p = 0.07$, $n = 8$).

Sinks act differently on content. Self Forcing's sink keeps more of it at 60 s ($\kappa$ 0.61 vs 0.14, +0.48 [0.39, 0.57], preregistered), but not steadily: $\kappa$ falls to 0.32 at 30 s and rises again, and the dunes, the sea and the mountains of the opening reappear after a minute of city (Fig. 3, left). The returns are positional (study 16, 24 openings): with the sink's keys re-rotated to sit just before the window, they vanish (burst contrast $-0.29$ [-0.37, -0.22]), and the content kept at 60 s falls from 0.56 to 0.25 ($-0.31$ [-0.37, -0.25]). With this one seed the re-rotated sink kept no detectably more content than no sink (+0.10 [-0.00, 0.21], not supported). Three new seeds per opening (study 21, 240 videos) reproduce the sink's advantage, the loss under re-rotation ($-0.40$ [-0.47, -0.34]) and the positional return, and, averaged per opening, detect a small share kept by the re-rotated sink: 0.19 against 0.11 without a sink (+0.08, Holm $p = 0.006$; [-0.01, 0.16] when the fresh starts

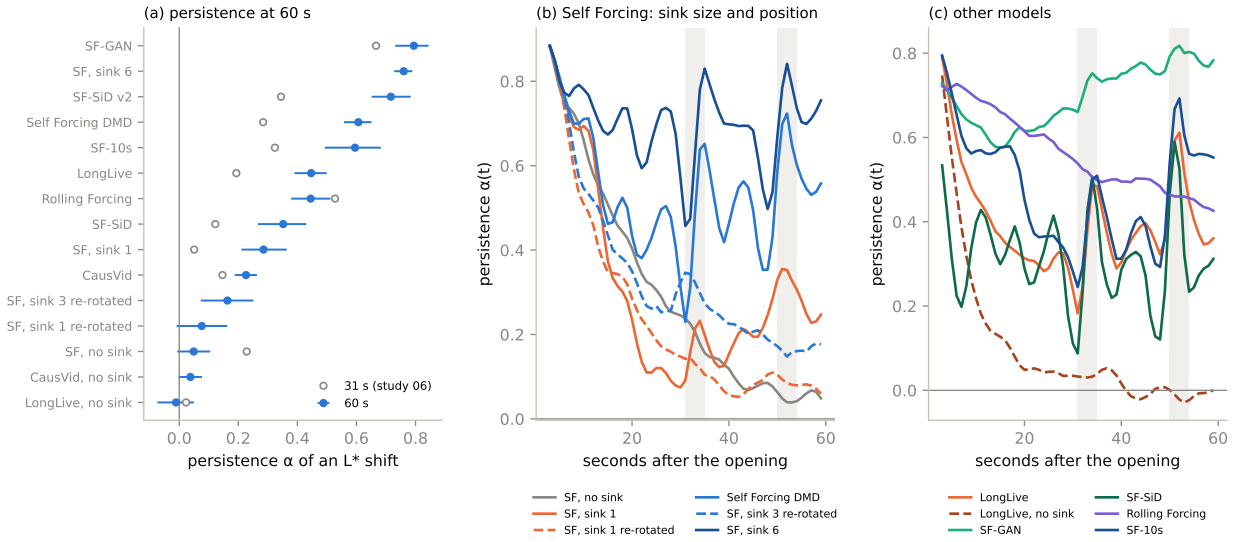



**Figure 2:** Persistence at 60 s (study 10; CausVid from study 7). (a) $\alpha$ of an L* shift with scene-bootstrap 95% CIs; hollow markers: 31 s (study 6). (b) Self Forcing with sinks of 0, 1, 3 and 6 latent frames, at their original positions or re-rotated to sit just before the window (dashed). (c) Other models. Curves: 1-s windows, 3-s moving average; shaded: 31–35 s and 50–54 s, where sinks of three or more frames at their original positions produce bursts. The 60-s endpoint window, [50, 60) s, overlaps the second burst; Appendix C reports a burst-free window.

are also resampled). What a Self Forcing sink keeps of the scene, it keeps mostly by periodically re-injecting the opening. LongLive's sink, which keeps tone, does not keep content (0.22 vs 0.24, preregistered test not supported).

Content relaxes outside the Wan-1.3B family too. FramePack, continuing the same 24 openings under the city prompt, keeps 0.55 of the class evidence at 20 s (Self Forcing: 0.40 with a sink, 0.31 without; study 17). Giving its two routes different images separates them (Appendix Fig. 6b): the clean context prefix alone keeps 0.24 [0.12, 0.37], levelling off after 14 s, while the SigLIP image embedding alone, with a city frame in the context, *pulls* the video toward the opening's class, to 0.23 [0.16, 0.30] at 20 s and still rising. The same embedding has no effect on tone (study 11), so in FramePack it sets the default of the scene but not of its brightness. [Krea 14B (study 22) and MAGI-1 (study 19) content results.]

## 4.2 The default

**The conditioning sets it.** Continuing the same 40 openings under four lighting prompts gives four defaults: on Self Forcing, sunny 43, overcast 34, dusk 19 and night 16 L*; on LongLive the same words give 45, 40, 32 and 21 (Appendix Fig. 7). The mapping from words to brightness is model-specific. Under one prompt the default also differs between checkpoints of one architecture (zero-drift levels from 17 to 54 L*), and it tracks where prompt-only rollouts end ($r = 0.93$, slope 1.58; Appendix C).

**Persistence depends on the scene.** Equation 1 separates a model term ($\alpha$) from a prompt term ($\mu$), but $\alpha$ itself varies: Self Forcing keeps 0.34 of the opening under a sunny prompt and 0.05 under a night prompt (exploratory interaction, $p = 0.001$). In four new scene types with their own prompts and self-generated openings (a park walk, a drone flight over a coast, a kitchen, a market; study 12), its persistence at 60 s differs between them (0.44 to 0.67, $p = 0.049$, preregistered), as does LongLive's at 30 s (0.10 to 0.43). The sink effect replicates almost exactly outside driving scenes: 0.55 with a sink and 0.02 without (+0.53 [0.45, 0.61]).

**Two components of drift.** For a prompt-only rollout, Eq. 1 says nothing about why the default should differ from where the same prompt starts. It does, and by how much depends on training (Appendix Fig. 5a).

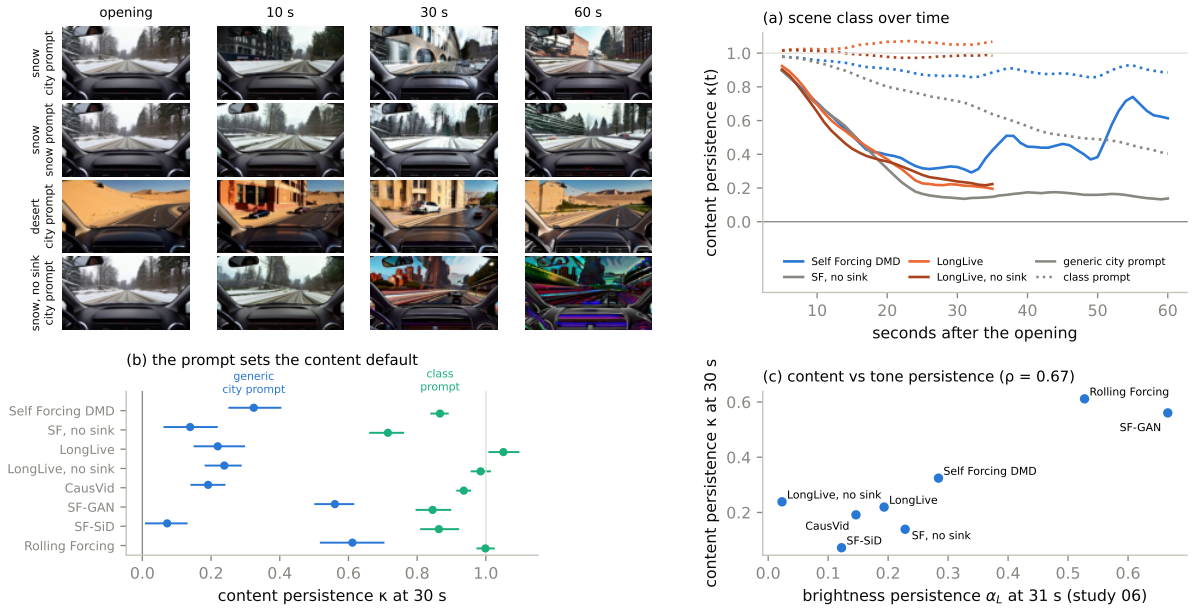



**Figure 3:** Scene content relaxes to the prompt's default (study 9). Left: a snow and a desert opening continued by Self Forcing DMD under the city prompt or their own class prompt. Under the city prompt both become city streets; with a sink the desert returns at 60 s; without a sink the video collapses. (a) Content persistence $\kappa(t)$ (CLIP log-odds of the opening's class against "city street", 0 = fresh generic rollout, 1 = opening) under the city prompt (solid) and the class prompt (dotted). (b) $\kappa$ at 30 s, 24 openings, 95% CIs. (c) Content against brightness persistence across configurations.

Fresh rollouts of Self Forcing (trained on its own 5-s rollouts) and CausVid end 4–11 L* from where they start; those of LongLive and Rolling Forcing, trained or tuned on their own long rollouts, 0.7–2.3 L*. Yet LongLive still forgets its opening ($\alpha = 0.02$ without a sink at 31 s). In this exploratory analysis drift has two separable parts: *forgetting* the opening, $(1-\alpha)(x_{\text{open}} - \mu)$, which every attention-based model does, and an *attractor offset* $\mu - x_{\text{start}}$, which is near zero in the two models trained on their own long rollouts (other differences between them and the short-clip models cannot be excluded). Two later models fit this pattern: fresh rollouts of Krea 14B, distilled with Self Forcing, brighten by 4.8 L* over 31 s (8 seeds), and those of MAGI-1, trained chunk by chunk on real videos rather than on its own rollouts, by 15–20 L* over 30 s. Prompt-only drift metrics mix the two, and both keep rising over 31 s even with a sink (Appendix H); relaxation describes the memory of the opening, not the absence of degradation.

# 5 What decides how much of the opening is kept

Figure 4 collects the long-run persistence of every model and configuration we ran for 60 s, grouped by how the model holds the opening. Four regimes appear. An opening seen only through a sliding key–value window is forgotten. Re-denoising overlapping frames passes it forward and delays forgetting by about 20 s. An opening kept next to the window, at positions of the kind the model was trained on, holds a steady share whose size depends on the model. A sink at its original rotary positions keeps the most, but largely through periodic positional bursts.

**Only in the window: forgotten.** Without a sink, Self Forcing and LongLive keep 0.05 and −0.01 at 60 s (study 10). MAGI-1 [31] (4.5B, distilled), which shares neither architecture nor training with the other models, attends in its released configuration to the four previous 1-s chunks, so the opening leaves its context after about 4 s; persistence falls from 0.87 at 5 s to 0.34 at 30 s (−0.52 [-0.70, -0.35], study 18) and to [x] at 60 s (study 19). Krea Realtime 14B [32], Self Forcing distilled from Wan2.1-14B and ten times the size of the other Wan models, recomputes its cache over a window of three latent frames (study 22, 24 scenes): it keeps

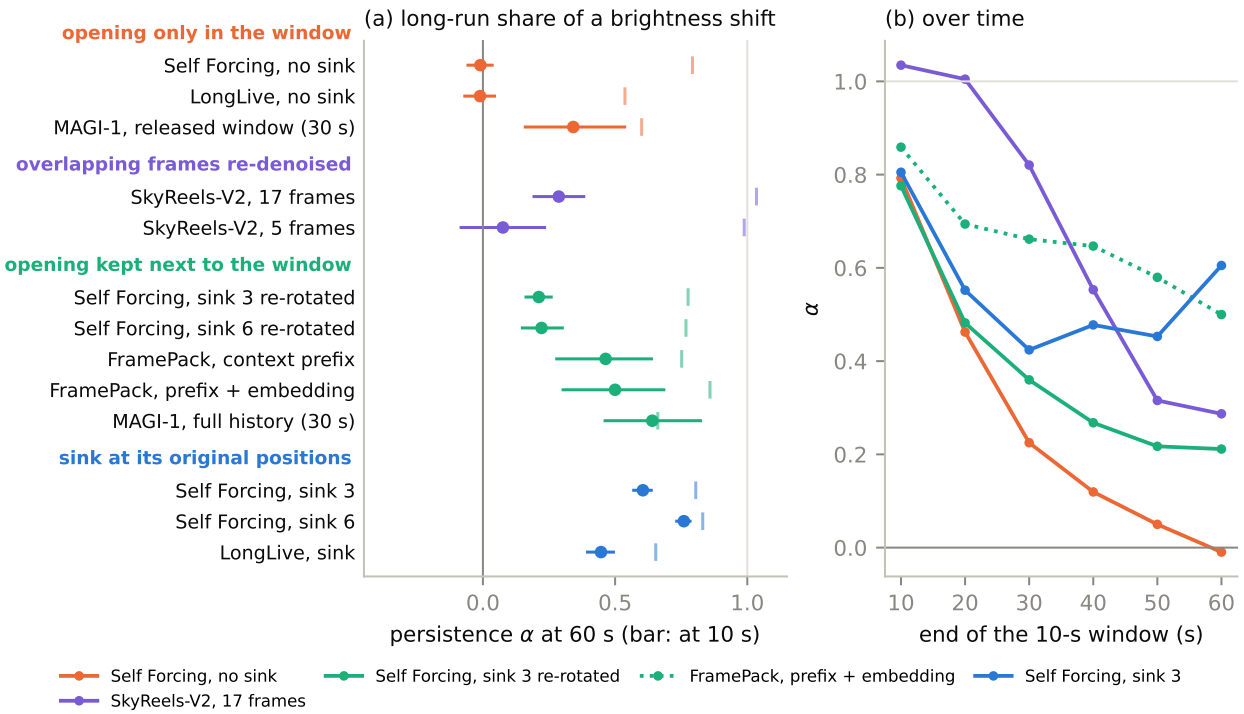



**Figure 4:** How the opening is held decides how much of it is kept. (a) Persistence of an L* shift at 60 s (window [50, 60) s; MAGI-1 full history: 30 s) with scene-bootstrap 95% CIs; ticks: at 10 s. (b) Persistence over time (10-s windows) for one configuration per regime.

0.33 at 10 s and nothing from 20 s on (−0.05 at 60 s). Scale does not change the regime.

**Sinks: depth helps only at the original positions.** In study 6, removing LongLive's sink lowered $\alpha$ at 31 s from 0.19 to 0.02 (−0.17 [-0.28, -0.06]); for Self Forcing the preregistered test failed at 31 s (−0.06 [-0.17, 0.05]) because without a sink it forgets slowly (Fig. 2b). At 60 s the difference is large (studies 7, 10): no sink 0.05, a sink of one latent frame 0.29, three 0.61, six 0.76 (+0.15 [0.11, 0.20] for six vs three), and 0.45 against −0.01 for LongLive. The sink's *position* matters more than its depth. Self Forcing keeps the sink keys at their absolute rotary positions, ever further from the current chunk; re-rotating them to sit just before the window, as Rolling Forcing and Deep Forcing do, lowers persistence at 60 s from 0.61 to 0.16 (−0.44 [-0.54, -0.35], preregistered; 40–50 s, between the bursts: −0.27 [-0.38, -0.16]). Re-rotated, depth no longer helps (study 16): sinks of six and three frames keep the same share (0.22 vs 0.23, −0.00 [-0.11, 0.10]; the preregistered depth test is not supported), and their curves coincide from 10 s on. Three new seeds per scene (study 20, 432 videos) reproduce this: 0.61 with a sink, 0.21 re-rotated and −0.01 without, the re-rotated sink keeping more than none (+0.22 [0.15, 0.29]), in each seed alone too. Krea's optional first-frame anchor, one latent frame kept next to the window, keeps 0.03 at 60 s ([test A3]), like a re-rotated one-frame sink on Self Forcing (0.08). A sink at in-distribution positions thus holds a small, steady share of tone, at most about 0.2 whatever its depth, and a smaller share of the scene (Section 4.1).

**Sinks at their original positions: bursts.** Persistence with such a sink jumps between 31.5 and 34.5 s (+0.45 [0.36, 0.53] with three frames, +0.46 with six) and again near 52 s, in LongLive and the other Self Forcing checkpoints too (Fig. 2b, c); the returns of the opening's scene coincide with these bursts. With re-rotated sinks the jump is replaced by a small decline (−0.08 [-0.13, -0.02]). The timing matches the "sink collapse" that LoL [21] attributes to rotary-position periodicity, and the role of intrinsic rotary frequencies in video length extrapolation [51, 52]: the sink keeps its absolute position, and at certain distances the current chunk attends to it strongly. Much of what a Self Forcing sink keeps at 60 s is therefore positional.

**Overlapping frames delay forgetting.** SkyReels-V2-DF extends videos in 97-frame segments that re-denoise the last 17 frames of the previous segment jointly with the new ones. It does not relax over 20 s ($\alpha = 0.96$ [0.86, 1.07]), whatever its guidance, history noise or number of steps (exploratory), and in 5 of 12 scenes $\alpha > 1$: the overlap passes the opening forward and can amplify it. Cutting the overlap to 5 frames makes it relax ($-0.24$ [-0.40, -0.09] at 15–20 s, preregistered). But the overlap only passes the opening forward: over 60 s (study 16) 17 frames hold $\alpha \approx 1$ until 20 s and then forget steadily, to 0.29 at 60 s ($-0.72$ [-0.81, -0.63] from 20 to 60 s), and 5 frames fall to 0.08 (both preregistered).

**A copy of the opening in every section is a permanent sink.** FramePack [23], a 13B model fine-tuned from HunyuanVideo [24], places the opening as a clean prefix frame in the context of *every* 33-frame section and also feeds its SigLIP embedding as a global condition (study 11, 12 scenes). It relaxes too: 0.91 of a brightness shift at 5 s and 0.68 [0.57, 0.79] at 20 s (preregistered), about as much as the best cache configurations over the same window (Appendix Table 4). The shift travels through the prefix: sent only through the prefix it keeps about half, only through the embedding nothing (0.01 [-0.14, 0.17]). Unlike the overlap, the prefix keeps its share: at 60 s (study 16) FramePack keeps 0.51 [0.29, 0.71] with both routes and 0.46 [0.28, 0.65] with the prefix alone, with little decline after 20 s ($-0.17$ [-0.38, 0.01] and $-0.08$ [-0.26, 0.09]; one collapsed scene excluded). MAGI-1 with full history, the opening never leaving the context, behaves alike: its persistence levels off near 0.65 from 10 s on (0.64 [0.45, 0.83] at 30 s, +0.30 [0.17, 0.43] over the released window). Keeping the opening does not stop the shared drift, though: MAGI-1's fresh rollouts brighten by 15–20 $L^*$ under either setting, so full history preserves the difference between openings, not their level.

**An action-conditioned world model.** Matrix-Game-2.0 [50] generates a world from an image and keyboard and mouse actions, with no text; the opening's latent leaves its 6-frame window after about a second, while its CLIP [48] embedding conditions every chunk (study 15). The model often collapses to black (40 of 144 videos within 30 s), which left 7 scenes for the preregistered 30-s analysis: a shift held only in the context faded ($-0.24$ [-0.35, -0.14], supported), and an embedding-only shift persisted (0.43 [0.13, 0.71]) without passing the Holm threshold ($p = 0.075$). At 15 s (20 scenes, exploratory) the context-only shift fades from 0.76 to 0.10 while the embedding-only shift grows from 0.08 to 0.25 [0.17, 0.36]: the rollout relaxes toward a default that moves with the conditioning (Appendix Fig. 6d).

**A phenomenological recursion.** Model a chunk's mean statistic as $x_{k+1} = w_w\, \bar{x}_{\text{window}} + w_s\, x_{\text{sink}} + w_p\, \mu$ with $w_w + w_s + w_p = 1$, where the opening counts as a window chunk until it rolls out and $\mu$ is the prompt's default. Without a sink the opening is eventually forgotten; with one, $\alpha_\infty = w_s/(w_s + w_p)$, so a small sink weight keeps a large share if the prompt's pull is also small. Fitted to 120-s curves, the recursion gives $w_s = 0.060$ [0.021, 0.097] and $w_w = 0.86$, implying $w_p \approx 0.08$. We preregistered this prediction and measured the prompt's pull by bit-exact replay of the key–value cache, regenerating one chunk from identical cache, noise and random state with only the text conditioning swapped to a lighting prompt with a different default (study 14): $w_p = 0.088$ [0.045, 0.149] at 15 s, proportional to the default difference (Appendix D). The recursion holds for about 20 s and no longer: with a sink the prompt's pull rises to 0.26 [0.17, 0.36] at 30 s (preregistered test of time invariance, +0.16 [0.10, 0.24]), and it explains neither the 0.2 kept by re-rotated sinks nor the 0.6 kept by aliased ones.

# 6 Two levers

Equation 1 suggests two ways to keep a long rollout near its opening without retraining: raise $\alpha$, or move $\mu$ toward the opening.

**Raising $\alpha$: a content-aware correction.** At denoising step 1 of 4, the noisy chunk is denoised once more with a temporary cache that holds only the anchor (the opening chunk). The difference between the per-channel spatial means of the two predictions (the DC tone) is added to the real prediction, scaled by $\beta$; the main cache and the random stream are untouched. In a preregistered confirmation on 40 held-out prompts × 4 seeds (study 2), colour drift $D$ fell by 1.48 [1.19, 1.76] at $\beta = 1$ and by 1.78 [1.45, 2.11] at $\beta = 1.5$, with structure drift $K$ non-inferior, and in the relaxation study the correction raised Self Forcing's persistence

**Table 1:** Correction versus post-hoc colour matching and deeper sinks (study 2 confirmation set: 40 held-out prompts × 4 seeds; transitions: 7 held-out prompts with legitimate brightness changes). Transitions kept: median over prompts of the corrected over the uncorrected first-to-last brightness change, prompt-bootstrap 95% CI.

| Method | $\Delta D$ [95% CI] | transitions kept | jitter | boundary jumps |
|---|---|---|---|---|
| Content-aware correction, $\beta = 1$ | $-1.48$ [-1.76, -1.19] | 0.83 [0.67, 1.47] | $\times 1.13$ | $\times 1.32$ |
| Content-aware correction, $\beta = 1.5$ | $-1.78$ [-2.11, -1.45] | 0.93 [0.83, 1.48] | $\times 1.67$ | $\times 2.40$ |
| Post-hoc mean matching | $-0.89$ [-1.29, -0.50] | 0.07 | $\times 0.15$ | $\times 0.12$ |
| Post-hoc Reinhard [20] | $-1.58$ [-1.96, -1.19] | 0.12 | $\times 0.13$ | $\times 0.10$ |
| Deeper sink, 6 frames (study 13) | $-1.30$ [-1.68, -0.93] | 0.70 [0.58, 1.12] | $\times 1.04$ | $\times 0.98$ |
| Deeper sink, 9 frames (study 13) | $-1.93$ [-2.31, -1.55] | 0.59 [0.19, 1.09] | $\times 1.16$ | $\times 1.09$ |

from 0.28 to 0.40 ($+0.12$ [0.05, 0.20], preregistered). Table 1 compares it with two baselines. Post-hoc colour matching to the anchor reaches a similar $D$ but, by construction, removes legitimate global changes too: on 7 held-out prompts with real brightness transitions (tunnels, garages, clouds), matching keeps 7–12% of the transition and the correction a median of 83% (per-prompt values from 0.58 to 1.71, so on some prompts it amplifies the change). A deeper attention sink, as in Deep Forcing [9] (study 13, bit-identical pairing), is a strong alternative: nine sink frames cut $D$ by more than the correction at $\beta = 1$ ($-0.45$ [-0.75, -0.15]) with smaller boundary jumps, but a sink pulls everything toward the opening, and in two of the seven transition prompts it erased most of a legitimate change (7% and 19% kept, against 151% and 147% for the correction). The correction costs one extra forward pass per chunk (about 25% more compute) and more frame-to-frame jitter; it is the more conservative choice when real changes matter. Correcting at step 0 removes as much drift but erases tunnels, and steps 2–3 do nothing, consistent with global tone being fixed at the earliest denoising steps [41, 42].

**Moving $\mu$: prompts.** Since the prompt sets the default, lighting words can be chosen so that the default matches the opening. Choosing the lighting sentence whose *measured* default is nearest the opening's brightness reduces $|\Delta L|$ at 31 s by 26% on Self Forcing ($-2.45$ [-4.08, -1.04], Holm $p = 0.013$, study 4). A free-form Qwen2.5-VL [18] caption of the opening does not reduce brightness drift (study 3; it calls half of the normally lit frames "dim"), and the other five preregistered prompt tests failed, including LongLive with its own calibration (study 5). The mapping from words to brightness is model-specific and must be measured per model; Appendix F shows how well Eq. 1 forecasts these outcomes.

# 7 Discussion

Any autoregressive sampler with finite memory that is ergodic under a fixed conditioning will eventually forget its initial state and settle into a conditioning-dependent stationary distribution, so that video rollouts relax is expected. What we measure is *how much* is kept and *what* decides it, and three answers hold across the models we tested. The long-run share depends on how the opening enters the computation: once the opening leaves a key–value window it is forgotten (in the Wan-based cache models at 1.3B and 14B parameters, and in MAGI-1), while an opening that stays in the context (a sink, MAGI-1's full history, FramePack's per-section prefix) holds a share. Re-denoising overlapping frames passes the opening forward and so delays forgetting by about 20 s, but does not stop it. A sink at in-distribution positions holds a small share of tone, about 0.2 whatever its depth, and less of the scene (0.08 more than none); the larger shares and the returning scenes of sinks at their original positions are positional bursts. The stationary state is set by the conditioning, for tone and for scene content, and models differ in which part of the conditioning carries which attribute: FramePack's image embedding sets the scene but not its brightness. And the stationary state of models trained on short clips is offset from where the same prompt starts; long-rollout training removes the offset but not the forgetting.

For world models, this means that a model conditioned on a real frame and a description drifts toward the description's default unless it re-conditions on the frame. Persistence at a stated horizon and the default with its offset are cheap to measure with our protocol and would make long-video models comparable where a single quality score does not. Anchors that keep the opening also pass its errors forward, and positional

bursts can bring a forgotten opening back. In Self Forcing the mechanism that keeps the opening's scene is exactly the unstable one: keeping the opening's content while forgetting its errors is the open problem.

**Limitations.** Six of the nine models share the Wan2.1-1.3B architecture and a seventh is a student of its 14B sibling; FramePack and MAGI-1 have other bases. Content relaxation was replicated on FramePack, but the prompt dependence, the recursion and the decomposition rest on Wan-based models, and the world model collapses after 20–30 s. Interventions shift global tone or select a scene class (CLIP scores are a coarse proxy for content); openings are held images; most configurations use one seed per scene (the Self Forcing sink results on tone and content replicate with three more); Holm correction is applied within studies only; the correction was not compared with Pathwise test-time correction or FreqForcing.

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# A Models, preregistered tests, deviations and withdrawn claims

**Table 2:** Configurations. Wan-based cache models generate 832×480 at 16 fps in chunks of 3 latent frames (0.75 s); window and sink sizes are in latent frames.

| Configuration | Base; training | Steps | Context of the opening |
|---|---|---|---|
| Self Forcing DMD, SiD, SiD v2, GAN [2, 5, 6] | Wan2.1-1.3B; self-rollout on 5-s clips | 4 | window 21; sink 0/1/3/6, original or re-rotated positions |
| Self Forcing 10 s | Wan2.1-1.3B; self-rollout on 10-s clips | 4 | window 21, sink 3 |
| CausVid [1] | Wan2.1-1.3B; teacher-forced DMD | 3 | window 21, sink 0/3 |
| LongLive [7] | Wan2.1-1.3B; long-rollout tuning | 4 | window 12, sink 0/3 |
| Rolling Forcing [8] | Wan2.1-1.3B; rolling joint denoising | 5 | anchor block + 21 |
| Krea Realtime 14B [32] | Wan2.1-14B; Self Forcing distillation | 4 | key–value cache recomputed from the last 3 latent frames; optional first-frame anchor (1 of the 3) |
| SkyReels-V2-DF [10] | Wan2.1-1.3B architecture; diffusion forcing; 544p, 24 fps | 30 + CFG | 17 (or 5) re-denoised overlapping frames |
| Matrix-Game-2.0 [50] | Wan2.1-1.3B architecture, no text; actions; 352×640, 25 fps | 3 | first latent frame only, window 6, no sink; CLIP image embedding |
| FramePack-F1 / I2V [23] | HunyuanVideo 13B; 480p, 30 fps | 25 | clean prefix in every section; SigLIP image embedding |
| MAGI-1 4.5B-distill [31] | own architecture (chunk-wise autoregressive diffusion transformer); 432×768, 24 fps, chunks of 1 s | 64, pipelined | released: window of 4 earlier chunks, no sink; or full history |

**Deviations from the preregistrations.**

- *Study 1* was frozen before analysis but after generation of its main videos had started, and its arms were not pinned to one GPU architecture: 62% of its nominal seed pairs ran on different architectures. Its preregistered Wald intervals are reported; the prompt-cluster bootstrap interval for the main contrast is wider ([-10.45, -5.39] for $D$) but leads to the same conclusion.
- *Study 3:* a preempted job resumed on another GPU type and split four LongLive openings across architectures. We regenerated the affected arms on the original architecture; conclusions are unchanged.
- *Study 6:* the ODE-initialised Self Forcing student was replaced by SiD v2 before generation (it collapses to blur after the opening). GPU pinning was replaced by an architecture guard that regenerates a task's outputs when a resumed job lands on another architecture; paired arms of each preregistered contrast always share a task. Exploratory configurations (sink 1; correction on LongLive and CausVid)

were added after freezing but before any data, from separate task lists, so their contrasts with the preregistered arms are not guaranteed to share an architecture. Prompt-only reference rollouts were renamed after generation; no analysis code changed.

- *Study 7* reuses the 24 scenes of study 6, including the 6 whose 120-s curves motivated it. Restricted to the 18 other scenes, the sink effect is +0.53 [0.42, 0.63] (Self Forcing) and +0.18 [0.13, 0.24] (CausVid).
- *Studies 10 and 11* were run on two non-preemptible 4-GPU jobs and A40-only array tasks instead of the preregistered all-queue arrays, because no large GPU was free on the shared queue; all their videos ran on A40 GPUs, so pairing is unaffected. The other studies ran as preregistered.

**Withdrawn or revised claims.** During the project we withdrew four claims after checks: that MUSIQ rewards collapsed videos (it is merely insensitive); that the sink stabilises the default (trend difference +0.5 [-2.4, 3.5]); that the model follows its context by 76% (true only for artificial shifts); and that early dynamics forecast the final brightness. The recursion's predicted Self Forcing plateau of 0.25–0.54 was missed (0.58 at 60 s).

# B Drift metrics for prompt-only rollouts

For rollouts without a held opening, drift is measured against 10 (Self Forcing) or 5 (LongLive) fresh 5-s generations of the same prompt. $D$ is the per-frame Mahalanobis distance of (Lab means, saturation, log sharpness) from the fresh frames' distribution, and $K$ the mean cosine distance of a frame's DINOv2 feature to its 5 nearest fresh-frame features, normalised by the fresh clips' own value; both are averaged over seconds 29–31. Transitions kept is the median over held-out transition prompts of the corrected over the uncorrected first-to-last brightness change; values above 1 mean the correction amplified the change.

# C Robustness

**Linearity.** For 12 of 14 configurations the persistence estimated from ±8 and from ±16 shifts differs by at most 0.09, with intervals including 0. SiD and the sink-1 configuration give near-zero slopes for small shifts (−0.10, −0.15) and positive ones for large shifts (0.18, 0.10); LongLive with a sink keeps a darkening better than a brightening (0.36 vs 0.04). The linear law is a first-order description.

**The 60-s endpoint and the positional bursts.** The preregistered 60-s window, 50–60 s, overlaps the second burst. Over 40–50 s, Self Forcing keeps 0.48 with a three-frame sink and 0.20 with a re-rotated one (difference −0.27 [-0.38, -0.16]); over the full period 35–60 s, 0.53 and 0.19. Sink depth is monotone in every window (40–50 s: no sink 0.08, one frame 0.23, three 0.48, six 0.63). Some curves are not monotone decays (GAN and SiD v2 lose part of the shift early and regain it), and Rolling Forcing is still declining at 60 s. In the 120-s subset of study 6 (6 scenes), Self Forcing with a sink keeps 0.54, 0.48 and 0.44 at 60, 90 and 118 s. No value we report is shown to be an asymptote; $x_\infty$ in Eq. 1 denotes the long-run tendency, and $\alpha$ is always reported with its horizon.

**Re-rotated sinks across studies.** A re-rotated three-frame sink keeps 0.16 at 60 s in study 10 and 0.23 in study 16 because 9 of the 24 scenes ran on another GPU architecture in study 16, which amounts to resampling them; all 72 videos repeated on the same architecture were bit-identical.

**The default and fresh rollouts.** Across 10 configurations, the zero-drift level correlates $r = 0.93$ with where prompt-only rollouts end, but the slope is 1.58 and the mean absolute difference 4.9 $L^*$ (largest for the GAN and SiD v2 checkpoints, whose zero-drift levels are extrapolated from slopes far from 0). Fresh rollouts give the direction and rough size of the default, not its exact value.

**Persistence depends on the prompt.** In the lighting experiment, the slope of the 31-s change on the opening differs between lighting prompts on Self Forcing (prompt × opening interaction, opening-clustered Wald test $p = 0.001$): persistence is 0.34 under sunny, 0.24 overcast, 0.17 dusk and 0.05 night prompts. On LongLive the interaction is weaker ($p = 0.053$). Dark defaults compress the range of brightness the model produces, which may explain part of this. Study 12 confirms the prompt dependence in four new scene types (Section 4.2).

**Content scores do not depend on the encoder.** Every content result uses CLIP ViT-L/14 class log-odds. We rescored all content videos with SigLIP so400m-patch14-384 [49], with its own text embeddings of the same seven labels, and reran each frozen analysis unchanged (Table 6). Every verdict agrees. Estimates in $\kappa$ agree within 0.03; raw log-odds differ in scale between encoders. This includes FramePack's embedding route, whose conditioning encoder is itself a SigLIP model: CLIP and SigLIP give it the same content share (0.23 and 0.25).

**Real and generated openings.** On Self Forcing, real and generated openings fall on the same line ($p = 0.80$); on LongLive, generated openings end 4.3 L* brighter than real ones of equal brightness ($p = 0.03$).

# D The recursion and the replay

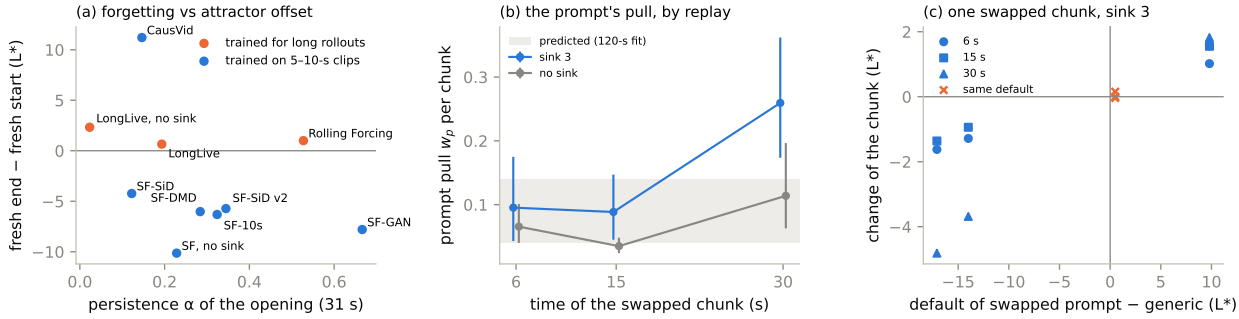



**Figure 5:** (a) Two components of drift: persistence of the opening (study 6) against the attractor offset of fresh rollouts (end minus start brightness, 4 seeds); models trained on their own long rollouts have almost no offset but still forget. (b) The prompt's per-chunk pull measured by bit-exact replay (study 14; 16 seeds; 95% CIs): one chunk regenerated from identical cache, noise and random state with only the text conditioning swapped; grey: the value implied by the recursion fitted to 120-s curves, preregistered for 15 s. (c) The swapped chunk's change against the difference of the prompts' defaults; a prompt with the same default has no effect.

Bit-exact replay of the key–value cache measures the recursion's weights directly, by regenerating one chunk from identical noise after changing only the sink, the window or the text conditioning: $w_s = 0.053$ $[0.040, 0.066]$ for synthetic tone shifts (0.036 in a dose–response design), and $w_w = 0.76$ (synthetic) to 0.90–0.99 (natural differences), in 10 development prompts × 2 seeds. The prompt's pull measured in study 14 is proportional to the difference between the two prompts' defaults, and a prompt with the same default has no effect (Fig. 5b, c). The three weights sum to 1.00 with the fitted $w_w$, but to 0.90 or 1.13 with the replay-measured ones, and $\alpha_\infty$ is 0.29–0.41 depending on which $w_s$ is used. Without a sink the measured pull is smaller (0.035 at 15 s), so $w_p$ is not a property of the prompt alone, and fitted weights depend on the horizon. With in-distribution positions a sink keeps 0.16–0.23 whatever its depth, below what the recursion implies; the aliased sink keeps 0.61, above it.

# E Models outside the Wan cache family

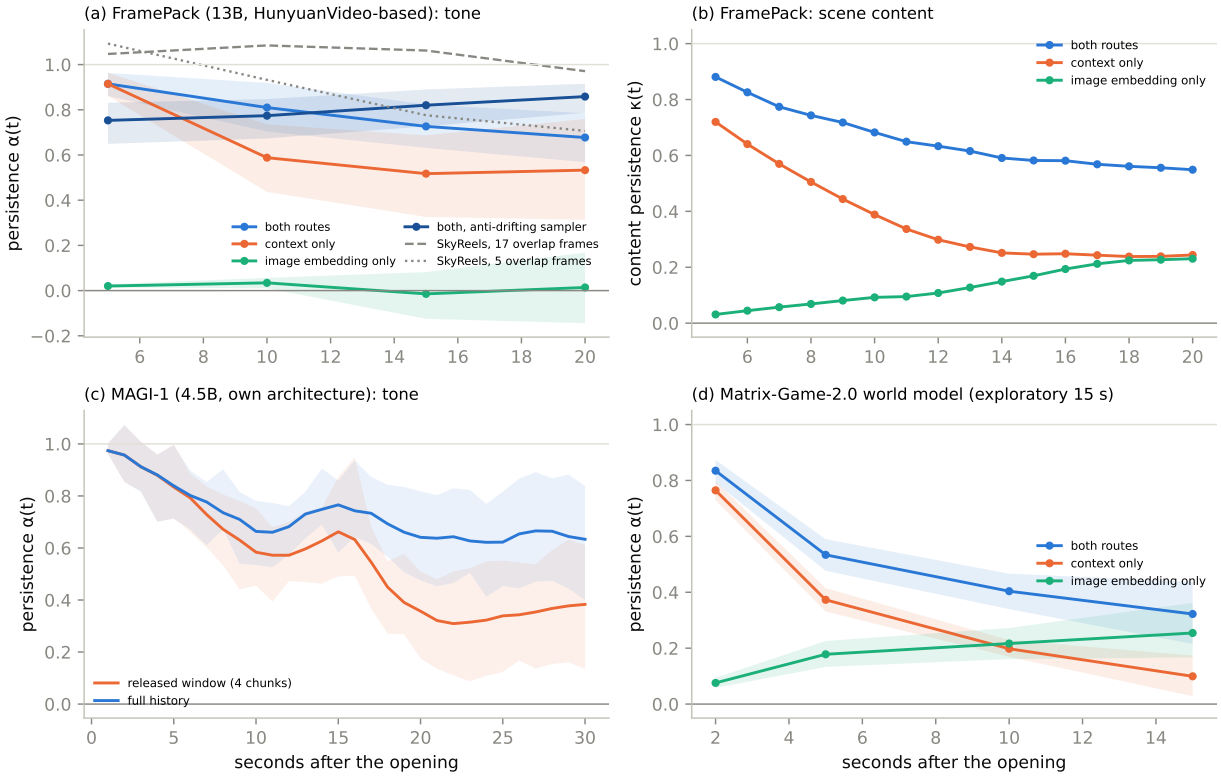



**Figure 6:** (a) FramePack (study 11, 12 scenes): the opening's brightness shift sent through both routes, only the clean context prefix, or only the SigLIP image embedding; the anti-drifting sampler; SkyReels-V2 with 17 and 5 overlapping frames for reference. (b) FramePack, scene content (study 17, 24 openings, city prompt): with a city frame in the context, the opening's embedding alone pulls the video toward the opening's class. (c) MAGI-1 (study 18, 12 scenes): the released window of four 1-s chunks against full history (1-s windows, 3-s moving average). (d) Matrix-Game-2.0, an action-conditioned world model without text (study 15, exploratory 15-s analysis, 20 scenes): the context latent leaves the window and is forgotten, while the rollout drifts toward the CLIP-embedded opening. Bands: scene-bootstrap 95% CIs.

# F The default and the forecast

**How well does the law forecast?** Taking $\alpha$ and the generic default from the relaxation study and the lighting defaults from the lighting experiment (different scenes and openings), $\Delta L \approx (1-\alpha)(\mu - L_{\text{open}})$ predicts the 31-s brightness change of each opening in the calibrated-lighting study with $r = 0.82$ (Self Forcing) and 0.77 (LongLive), mean absolute error 3.8 L$^*$. Most of this correlation reflects the opening's brightness. The forecast of the *benefit* of calibration per opening is weaker ($r = 0.46$ and 0.58), and its mean is overpredicted ($-4.5$ predicted vs $-3.2$ and $-2.8$ observed L$^*$). The law is a first-order guide to a prompt choice, not a precise forecast.

# G Pitfalls for evaluating long-video models

1. **The GPU architecture changes the video.** With bf16, outputs are bit-identical within an architecture and differ across A40, A100/A800 and L40S/RTX 6000 Ada. Paired designs must pin or guard the architecture.
2. **Resumed jobs can silently break pairing.** Generators that skip existing outputs resume a preempted job wherever it lands.
3. **Image-quality scores miss drift.** MUSIQ, used by VBench for imaging quality, varies only between

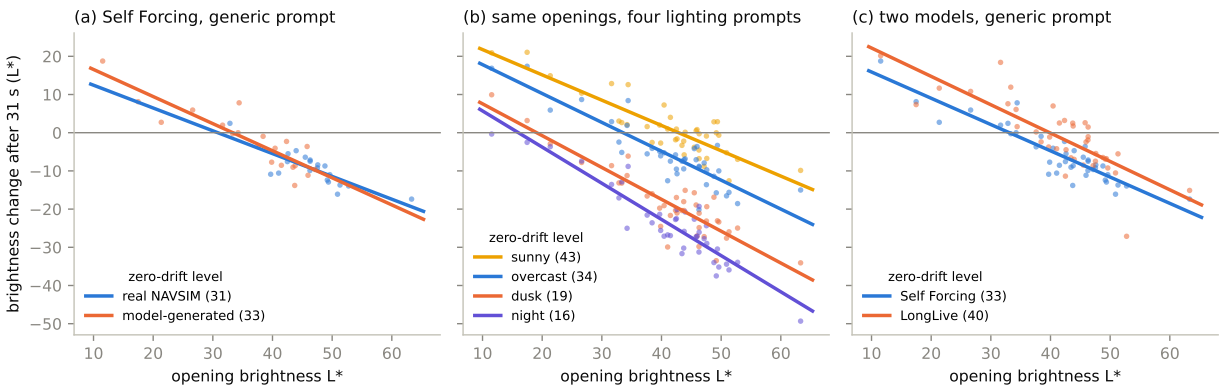



**Figure 7:** Brightness change after 31 s against the opening's brightness (one dot per opening, generic or lighting prompt): (a) real and generated openings follow one line; (b) four lighting prompts set four defaults (Self Forcing); (c) two models: similar slopes, different defaults.

66 and 69 across context strategies whose colour drift differs threefold.

4. **Natural variation is too small to estimate persistence.** Within a prompt, seeds start only about 2 $L^*$ apart, so within-prompt slopes of end on start brightness have intervals of about $\pm 0.4$; controlled shifts are needed.
5. **Change-on-baseline regressions mislead.** Regressing change on a baseline measured with error produces spurious regression to the mean [46, 47]. Held openings are measured almost without error, and the interventional design removes the issue.

# H Context strategies (study 1)

A preregistered factorial study of 10 window and sink strategies on Self Forcing (40 prompts $\times$ 4 seeds, 31 s) found that a 3-frame sink cuts colour drift $D$ by 7.56 [6.81, 8.32] and DINOv2 drift $K$ by 0.90; that window size matters only without a sink; and that re-rotating the sink's positions, which Self Forcing was not trained for, does not reduce $D$ ($-0.03$ [-0.79, 0.72]) and increases $K$ ($+0.36$). Darkening persists with a sink (last-2-s $z_L = -1.99$). $D$ and $K$ rise roughly linearly over 31 s under every strategy, without a visible plateau.

**Table 3:** All preregistered primary tests and their outcomes. Intervals are Wald intervals from the preregistered mixed models in studies 1, 2 and 13 and bootstrap intervals otherwise; $p$ is Holm-adjusted within each study (no correction across studies). Study 3 LongLive rows are as executed; after regenerating four cross-architecture openings the conclusions are unchanged. Units: $D$, $K$ drift metrics; $|\Delta L|$ in L$^*$; $\alpha$ persistence, $\kappa$ content persistence. Arms: G generic prompt, M VLM caption of the opening, C calibrated lighting plus caption, ttc content-aware correction. [‡]The $t$-test treats the default $\mu$ as fixed; the bootstrap interval also resamples the fresh starts that estimate it (pooled with seed 0: $[0.02, 0.15]$).

| Study | Test | Model | Estimate [95% CI] | $p_{\text{Holm}}$ | Outcome |
|---|---|---|---|---|---|
| 1 Context | sink 3 vs none, $D$ | SF | $-7.56$ [-8.32, -6.81] | $< 0.001$ | supported |
| | sink 3 vs none, $K$ | SF | $-0.90$ [-1.04, -0.76] | $< 0.001$ | supported |
| | window × sink, $D$ | SF | $+5.71$ [4.21, 7.22] | $< 0.001$ | supported |
| | window × sink, $K$ | SF | $+0.67$ [0.38, 0.95] | $< 0.001$ | supported |
| 2 Correction | ttc $\beta$=1, $D$ | SF | $-1.48$ [-1.76, -1.19] | $< 10^{-23}$ | supported |
| | ttc $\beta$=1.5, $D$ | SF | $-1.78$ [-2.11, -1.45] | $< 10^{-24}$ | supported |
| 3 Caption | M − G, $\|\Delta L\|$ | SF | $-1.21$ [-3.09, 0.36] | 0.58 | not supported |
| | M+ttc − M, $\|\Delta L\|$ | SF | $-1.00$ [-1.89, -0.12] | 0.14 | not supported |
| | M − G, $\|\Delta L\|$ | LL | $-0.19$ [-2.00, 1.63] | 1.0 | not supported |
| | M+ttc − M, $\|\Delta L\|$ | LL | $-0.36$ [-1.48, 0.75] | 1.0 | not supported |
| 4 Calibrated | C − M, $\|\Delta L\|$ | SF | $-1.24$ [-2.54, 0.12] | 0.25 | not supported |
| | C − G, $\|\Delta L\|$ | SF | $-2.45$ [-4.08, -1.04] | 0.013 | supported |
| | C − M, $\|\Delta L\|$ | LL | $-1.15$ [-2.59, 0.29] | 0.26 | not supported |
| | C − G, $\|\Delta L\|$ | LL | $-1.34$ [-3.19, 0.55] | 0.26 | not supported |
| 5 Calibrated (LL scale) | C − M, $\|\Delta L\|$ | LL | $-1.36$ [-3.06, 0.32] | 0.26 | not supported |
| | C − G, $\|\Delta L\|$ | LL | $-1.55$ [-3.50, 0.39] | 0.26 | not supported |
| 6 Relaxation | no sink − sink, $\alpha_L$ (31 s) | SF | $-0.06$ [-0.17, 0.05] | 0.34 | not supported |
| | no sink − sink, $\alpha_L$ (31 s) | LL | $-0.17$ [-0.28, -0.06] | 0.013 | supported |
| | ttc − none, $\alpha_L$ (31 s) | SF | $+0.12$ [0.05, 0.20] | 0.013 | supported |
| | no sink − sink, $\alpha_b$ (31 s) | SF | $-0.26$ [-0.34, -0.18] | $< 0.001$ | supported |
| 7 Sink at 60 s | sink − no sink, $\alpha_L$ | SF | $+0.55$ [0.45, 0.65] | $< 10^{-9}$ | supported |
| | sink − no sink, $\alpha_L$ | CausVid | $+0.19$ [0.14, 0.24] | $< 10^{-6}$ | supported |
| 8 Overlap | 5 − 17 frames, $\alpha_L$ (20 s) | SkyReels | $-0.24$ [-0.40, -0.09] | 0.014 | supported |
| 9 Content | class − city prompt, class score (30 s) | SF | $+9.33$ [8.11, 10.51] | $< 10^{-9}$ | supported |
| | sink − no sink, $\kappa$ (30 s) | LL | $-0.02$ [-0.10, 0.07] | 0.72 | not supported |
| | sink − no sink, $\kappa$ (60 s) | SF | $+0.48$ [0.39, 0.57] | $< 10^{-7}$ | supported |
| 10 Atlas | sink 6 − sink 3, $\alpha_L$ (60 s) | SF | $+0.15$ [0.11, 0.20] | $< 10^{-6}$ | supported |
| | sink 1 − no sink, $\alpha_L$ (60 s) | SF | $+0.24$ [0.14, 0.34] | $< 0.001$ | supported |
| | sink − no sink, $\alpha_L$ (60 s) | LL | $+0.46$ [0.39, 0.53] | $< 10^{-10}$ | supported |
| | re-rotated − original sink, $\alpha_L$ (60 s) | SF | $-0.44$ [-0.54, -0.35] | $< 10^{-7}$ | supported |
| 11 FramePack | $\alpha$ (both routes, 20 s) > 0.5 | FramePack-F1 | 0.68 [0.57, 0.79] | 0.033 | supported |
| | $\alpha$ (embedding only, 20 s) > 0 | FramePack-F1 | 0.01 [-0.14, 0.17] | 0.87 | not supported |
| | both − context only, $\alpha$ (20 s) | FramePack-F1 | $+0.14$ [-0.05, 0.33] | 0.38 | not supported |
| 12 Prompts | sink − no sink, $\alpha_L$ (60 s), 4 domains | SF | $+0.53$ [0.45, 0.61] | $< 10^{-10}$ | supported |
| | sink − no sink, $\alpha_L$ (30 s), 4 domains | LL | $+0.19$ [0.11, 0.28] | 0.005 | supported |
| | $\alpha_L$ differs between domains (ANOVA) | SF | $F = 3.12$ | 0.049 | supported |
| 13 Deep sink | sink 6, $D$ | SF | $-1.30$ [-1.68, -0.93] | $< 10^{-6}$ | supported |
| | sink 9, $D$ | SF | $-1.93$ [-2.31, -1.55] | $< 10^{-6}$ | supported |
| 14 Prompt pull | $w_p > 0$ at 15 s (replay) | SF | $+0.088$ [0.045, 0.149] | 0.006 | supported |
| 15 World model | ctx-only shift fades (30 vs 5 s) | Matrix-Game | $-0.24$ [-0.35, -0.14] | 0.024 | supported |
| | emb-only shift persists, $\alpha$ (30 s) | Matrix-Game | $+0.43$ [0.13, 0.71] | 0.075 | not supported |
| | both − ctx-only, $\alpha$ (30 s) | Matrix-Game | $+0.33$ [-0.07, 0.71] | 0.18 | not supported |
| 16 Re-rotation | re-rotated − original, burst contrast $R$ | SF | $-0.29$ [-0.37, -0.22] | $< 10^{-5}$ | supported |
| | re-rotated − original sink, $\kappa$ (60 s) | SF | $-0.31$ [-0.37, -0.25] | $< 10^{-6}$ | supported |
| | re-rotated sink − none, $\kappa$ (60 s) | SF | $+0.10$ [-0.00, 0.21] | 0.11 | not supported |
| | re-rotated − original sink 6, $\alpha_L$ (60 s) | SF | $-0.54$ [-0.62, -0.45] | $< 10^{-9}$ | supported |
| | re-rotated sink 6 − sink 3, $\alpha_L$ (60 s) | SF | $-0.00$ [-0.11, 0.10] | 0.94 | not supported |
| | $\alpha$ (both routes, 60 s) > 0 | FramePack-F1 | 0.51 [0.29, 0.71] | 0.004 | supported |
| | $\alpha$ (context only, 60 s) > 0 | FramePack-F1 | 0.46 [0.28, 0.65] | 0.003 | supported |
| | 5 frames: $\alpha$ (60 s) − $\alpha$ (20 s) | SkyReels | $-0.57$ [-0.79, -0.39] | 0.001 | supported |
| | 17 frames: $\alpha$ (60 s) − $\alpha$ (20 s) | SkyReels | $-0.72$ [-0.81, -0.63] | $< 10^{-6}$ | supported |
| 17 FP content | content falls, class score (20 s) | FramePack-F1 | $-6.72$ [-8.29, -5.20] | $< 10^{-5}$ | supported |
| | $\kappa$ (context only, 20 s) > 0 | FramePack-F1 | 0.24 [0.12, 0.37] | 0.001 | supported |
| | $\kappa$ (embedding only, 20 s) > 0 | FramePack-F1 | 0.23 [0.16, 0.30] | $< 10^{-6}$ | supported |
| | both − context only, $\kappa$ (20 s) | FramePack-F1 | $+0.31$ [0.20, 0.41] | $< 0.001$ | supported |
| 18 MAGI-1 | window: $\alpha$ (30 s) − $\alpha$ (5 s) | MAGI-1 | $-0.52$ [-0.70, -0.35] | $< 0.001$ | supported |
| | full history − window, $\alpha$ (30 s) | MAGI-1 | $+0.30$ [0.17, 0.43] | 0.002 | supported |
| | $\alpha$ (full history, 30 s) > 0 | MAGI-1 | 0.64 [0.45, 0.83] | $< 0.001$ | supported |
| 20 Seeds | sink 3 − none, $\alpha$ (60 s) | SF | $+0.62$ [0.56, 0.67] | $< 10^{-15}$ | supported |
| | re-rotated − original sink, $\alpha$ (60 s) | SF | $-0.39$ [-0.46, -0.32] | $< 10^{-8}$ | supported |
| | re-rotated sink − none, $\alpha$ (60 s) | SF | $+0.22$ [0.15, 0.29] | $< 10^{-5}$ | supported |
| 21 Content seeds | sink 3 − none, $\kappa$ (60 s) | SF | $+0.48$ [0.40, 0.56] | $< 10^{-10}$ | supported |
| | re-rotated − original sink, $\kappa$ (60 s) | SF | $-0.40$ [-0.47, -0.34] | $< 10^{-12}$ | supported |
| | re-rotated sink − none, $\kappa$ (60 s) | SF | $+0.08$ [-0.01, 0.16][‡] | 0.006 | supported |
| | re-rotated − original, burst contrast $R$ | SF | $-0.34$ [-0.38, -0.28] | $< 10^{-10}$ | supported |

**Table 4:** Persistence over the same window, 15–20 s after the opening, for every model (exploratory; scene-bootstrap 95% CIs; Matrix-Game: scenes without a black second before 20 s).

| Configuration | $\alpha$ | 95% CI | scenes |
|---|---|---|---|
| SkyReels-V2-DF, 17 overlap frames | 1.01 | [0.89, 1.15] | 12 |
| FramePack-I2V, anti-drifting sampler | 0.86 | [0.79, 0.92] | 12 |
| SkyReels-V2-DF, 5 overlap frames | 0.77 | [0.64, 0.90] | 12 |
| Self Forcing DMD, sink 6 | 0.72 | [0.65, 0.78] | 24 |
| FramePack-F1, both routes | 0.68 | [0.56, 0.78] | 12 |
| Rolling Forcing | 0.64 | [0.60, 0.67] | 24 |
| Self Forcing GAN | 0.61 | [0.54, 0.67] | 24 |
| Self Forcing 10-s | 0.54 | [0.46, 0.63] | 24 |
| FramePack-F1, context prefix only | 0.53 | [0.31, 0.76] | 12 |
| Self Forcing DMD, sink 3 | 0.49 | [0.44, 0.53] | 24 |
| Self Forcing DMD, no sink | 0.44 | [0.38, 0.51] | 24 |
| Self Forcing DMD, sink 3 re-rotated | 0.38 | [0.28, 0.46] | 24 |
| Matrix-Game-2.0, both routes | 0.37 | [0.22, 0.52] | 17 |
| Self Forcing SiD | 0.34 | [0.24, 0.46] | 24 |
| LongLive, sink 3 | 0.33 | [0.25, 0.40] | 24 |
| Self Forcing DMD, sink 1 re-rotated | 0.33 | [0.25, 0.41] | 24 |
| Self Forcing DMD, sink 1 | 0.29 | [0.17, 0.41] | 24 |
| Self Forcing SiD v2 | 0.24 | [0.11, 0.36] | 24 |
| Matrix-Game-2.0, embedding only | 0.22 | [0.06, 0.36] | 17 |
| CausVid, sink 3 | 0.21 | [0.16, 0.26] | 24 |
| Matrix-Game-2.0, context only | 0.10 | [-0.02, 0.22] | 17 |
| LongLive, no sink | 0.08 | [0.04, 0.12] | 24 |
| CausVid, no sink | 0.08 | [0.04, 0.12] | 24 |
| FramePack-F1, embedding only | 0.01 | [-0.14, 0.17] | 12 |

**Table 5:** Every global statistic relaxes toward a default (observational; 40 held openings, generic prompt, 31 s). Persistence = 1 + slope of the change on the opening value; $r$ = correlation of the change with the opening value.

| | Self Forcing | | LongLive | |
|---|---|---|---|---|
| Statistic | persistence | $r$ | persistence | $r$ |
| $L^*$ | 0.31 | −0.91 | 0.26 | −0.81 |
| $a^*$ | 0.38 | −0.92 | 0.39 | −0.90 |
| $b^*$ | 0.22 | −0.88 | 0.42 | −0.80 |
| saturation | 0.54 | −0.92 | 0.23 | −0.95 |
| log sharpness | 0.37 | −0.91 | 0.23 | −0.94 |
| $L^*$ contrast | 0.34 | −0.82 | 0.37 | −0.79 |

**Table 6:** Content results do not depend on the scoring encoder (exploratory). Each preregistered content test rerun with its frozen analysis on SigLIP so400m embeddings instead of CLIP ViT-L/14. Estimates in each encoder's log-odds or $\kappa$ units; Holm $p$ within each study; $^\dagger$ not supported. The verdict agrees for 14 of 14 tests.

| | | CLIP | | SigLIP | |
|---|---|---|---|---|---|
| Study | Test | est. | $p$ | est. | $p$ |
| 09 H1 | prompt sets content: $s_{\text{class}} - s_{\text{generic}}$ (30 s) | +9.33 | <0.001 | +13.24 | <0.001 |
| 09 H2 | LongLive sink keeps content: $\Delta\kappa$ (30 s) | -0.02 | 0.718$^\dagger$ | +0.00 | 0.920$^\dagger$ |
| 09 H3 | Self Forcing sink keeps content: $\Delta\kappa$ (60 s) | +0.48 | <0.001 | +0.48 | <0.001 |
| 16 A1 | late return is positional: $\Delta R$ | -0.29 | <0.001 | -0.28 | <0.001 |
| 16 A2 | re-rotation lowers content: $\Delta\kappa$ (60 s) | -0.31 | <0.001 | -0.32 | <0.001 |
| 16 A3 | re-rotated sink keeps content: $\Delta\kappa$ (60 s) | +0.10 | 0.113$^\dagger$ | +0.09 | 0.099$^\dagger$ |
| 17 H1 | FramePack content relaxes: $s - s_{\text{open}}$ | -6.72 | <0.001 | -10.10 | <0.001 |
| 17 H2 | context prefix carries content: $\kappa$ | +0.24 | 0.001 | +0.22 | <0.001 |
| 17 H3 | SigLIP embedding carries content: $\kappa$ | +0.23 | <0.001 | +0.25 | <0.001 |
| 17 H4 | embedding adds content: $\Delta\kappa$ | +0.31 | <0.001 | +0.30 | <0.001 |
| 21 H1 | sink keeps content, 3 new seeds: $\Delta\kappa$ (60 s) | +0.48 | <0.001 | +0.50 | <0.001 |
| 21 H2 | re-rotation lowers content: $\Delta\kappa$ (60 s) | -0.40 | <0.001 | -0.40 | <0.001 |
| 21 H3 | re-rotated sink keeps content: $\Delta\kappa$ (60 s) | +0.08 | 0.006 | +0.10 | <0.001 |
| 21 H4 | late return is positional: $\Delta R$ | -0.34 | <0.001 | -0.32 | <0.001 |